european actuarial academy

The European knowledge centre for actuaries

Hands-on Porting

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Prepare the Working Repository

01
Hands-on Repo Clone

- Clone the repository in the VS Code and work offline
- Or: Work in the GitHub Codespaces:

  Code → Codespaces → Create codespace on main

02
Check Setup

- After starting, check the Setup:

```
pytest -s
```

- Expected:
  → no tests ran in 0.02s = Setup correct, as new
- An later after the exercise:
  → all tests pass = Result correct

Different Repositories to Be Cloned

01
Crafted Approach (`-ca`)

```
git clone https://github.com/bartlmac/portxlpy-seminar-2026-02-eaa-ex-ca.git

cd portxlpy-seminar2026-02-eaa-ex-ca
```

02
Industrial Approach (`-ia`)

```
git clone https://github.com/bartlmac/portxlpy-seminar-2026-02-eaa-ex-ia.git

cd portxlpy-seminar2026-02-eaa-ex-ia
```

Hands-on Porting Session: Industrial Approach (1)

Industrial Approach is the basis for agentic AI setup

Methodology

- Complete Excel-to-text extraction (formulas, VBA, structures)
- LLM-Pipeline: Automated generation with Pytest-Gates
- Golden Master comparisons and Unit-Tests
- CI/CD integration with version control of each iteration

Advantages

- Scalable and acceleratable
- Reproducible approach, transparency, and auditability
- Human-in-the-Loop remains necessary but maximally efficient

Risks & Limits

- Initial setup effort for Pipeline
- API complexity (Excel, VBA, LLM)
- Requires DevOps expertise

Goal: Execute one of the steps in the industrial approach

The LLM must receive the current state of knowledge.

Step 1

Open ChatGPT In The Browser

Step 2

Entry Prompt:

```
"Du steigst in ein laufendes Projekt ein …"
```

Step 3

Upload The Following Files From The Repo:

- protokoll.txt
- excelzell.csv
- excelber.csv
- …